

SAMPLE QUESTION PAPER

IB DIPLOMA

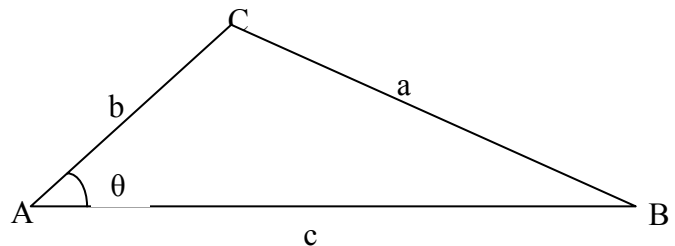
MATHEMATICS

1. Identify irrational numbers from the following. $\sqrt{2}, \pi, \frac{3}{2}, -3, 0, \sqrt{5}, 7, \sqrt{7}$.

2. Simplify $\sqrt{\frac{3}{2}} + \sqrt{\frac{2}{3}}$ in $a\sqrt{b}$ form and write the value of a and b.

3. Find the quadratic equation whose one of the roots is $2 - \sqrt{3}$..

4. In the given right triangle the longest side is c . Find the value of $\sin^2 \theta + \cos^2 \theta$.

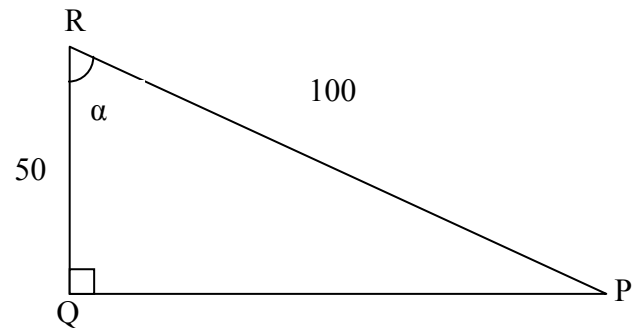


5. Find the area of a rhombus whose sides are 4cm and one of its diagonal is 6cm.

6. Define (i) Ortho centre (ii) circum centre.

7. If $(x-1)$ is a factor of the polynomial $P(x) = 2x^3 - ax^2 + 3x - 1$, then find the value of a.

8. Find the angle α in the adjoining figure:.



9. Simplify : $\frac{x^3 - 8}{x^2 + 2x + 4}$.

10. Which statistical graph is most appropriate to represent the following data? Illustrate.

Name of the cricketer	Sachin	Dhoni	Dravid	Pathan
Number of runs scored	50	40	70	65

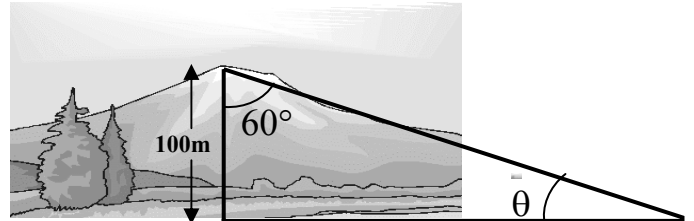
11. How many different numbers can be formed by arranging the digits 6,9,2 without repetition of any digit? How many of them are divisible by 2?

12. A cylinder of height 6 cm and radius 4 cm is cut along its height into two equal halves. Find the area of rectangle so formed.

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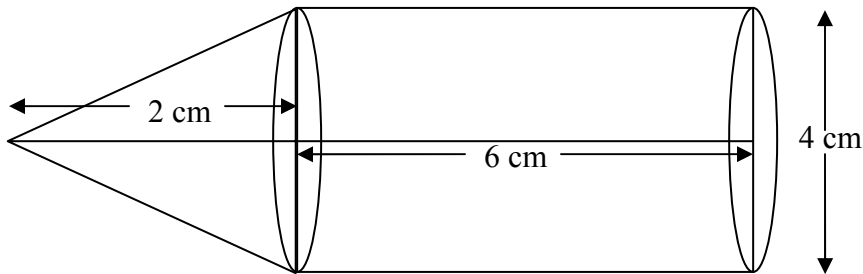
13. A circle of radius 3 cm is revolved along its diameter. Name the object so formed by this rotation and calculate its surface area.

14. Find the distance of point A from the top of the mountain in the given figure.



15. Find the volume of a circular cake whose height is 15 cm and area of its base is 30 cm^2 ..

16. Find the volume of the given figure and write your answer in terms of π ..



17. An IQ test on 28 students of class X reveals the following data

IQ	30-40	40-50	50-60	60-70	70-80
Frequency	3	7	10	2	6

Find the mean of the above data.

ALL THE BEST